BIAS SIMILARITY MEASUREMENT: A BLACK-BOX AUDIT OF FAIRNESS ACROSS LLMS

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ABSTRACT

Conventional fairness evaluations grade large language models (LLMs) in isolation, obscuring how biases align, diverge, or persist across families and releases. We introduce Bias Similarity Measurement (BSM), a black-box framework that audits fairness relationally by comparing models' bias signatures across scalar, distributional, behavioral, and representational views. BSM constructs pairwise similarity matrices using six signals: accuracy, bias scores under ambiguous and disambiguated contexts, output histograms, cosine similarity of output distributions, abstention (Unknown) flip rates, and CKA-based representational similarity. Evaluating a broad set of LLMs spanning LLaMA, Gemma, GPT, and Gemini on BBQ, UnQover, and open-ended StereoSet, we find: (i) instruction tuning often enforces refusal under ambiguity, masking bias that reappears under forced choice; (ii) representational similarity between base and tuned variants remains high (CKA diagonal 0.94+; full 0.85+); and (iii) small models can regress in fairness under forced choice. For example, LLaMA 3.1 8B drops ambiguous bias from 18.59 to 1.38 and disambiguated bias from 31.37 to 4.78, while LLaMA 3.2 3B increases disambiguated bias from 17.67 to 30.97. Meanwhile, GPT-4o Mini and GPT-5 Mini remain near-neutral (e.g., ambiguous bias 0.47 and 0.21). Collectively, these findings support auditing fairness as comparative bias similarity, not isolated scores, to distinguish refusal policies from genuine representational debiasing.

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1 INTRODUCTION

Fairness evaluation of LLMs increasingly determines deployment, procurement, and governance decisions. Yet most evaluations report model-specific scores, making it difficult to tell whether different models encode similar biases, whether new releases truly debias, or whether they merely refuse to answer sensitive questions. This paper argues that fairness should be audited relationally: we seek to understand how models compare in their bias behavior and representations, within and across families.

We propose Bias Similarity Measurement (BSM), a black-box framework that unifies evidence from four complementary views—scalar, distributional, behavioral, and representational—into a single similarity space. Concretely, BSM compares models along six signals: accuracy, bias scores, output histograms, cosine similarity, abstention flip rates, and CKA-based representational similarity (Kornblith et al., 2019). By separating ambiguous (abstention-appropriate) from disambiguated (answer-required) contexts in BBQ (Parrish et al., 2022), and by complementing with UnQover (Li et al., 2020) and StereoSet (Nadeem et al., 2021), BSM disentangles fairness via caution from fairness via representation.

Contributions. (1) We introduce BSM, a unified, black-box framework to measure comparative fairness via similarity. (2) We implement BSM across established bias benchmarks (BBQ, UnQover, StereoSet) and evaluate diverse model families, including LLaMA (Touvron et al., 2023; Dubey et al., 2024), Gemma (Team et al., 2024a;b), Gemini (Google AI Developers, 2025), and GPT variants (OpenAI, 2024; 2025). (3) We report relational findings and release a practical auditing workflow that supports procurement, regression testing, and lineage screening.

2 Related Works

Bias benchmarks for LLMs. The Bias Benchmark for QA (BBQ) (Parrish et al., 2022) delineates ambiguous and disambiguated contexts to expose bias under uncertainty versus evidence. UnQover (Li et al., 2020) probes stereotyping via underspecified questions using forced choice, and StereoSet (Nadeem et al., 2021) evaluates open-ended stereotypical associations. BSM builds on these ideas but shifts emphasis from isolated scores to relational comparisons across models and releases.

Similarity measures. BSM combines functional similarity (outputs and behavior) with representational similarity. For representations, we adopt centered kernel alignment (CKA) (Kornblith et al., 2019), chosen for robustness across independently trained networks. Recent work explores representational similarity for LLMs (Klabunde et al., 2023) and surveys functional/representational measures (Klabunde et al., 2025). Our contribution is to integrate these strands with fairness-specific behavioral signals (e.g., abstention flips) into a single auditing framework.

Model families. We study open and proprietary families spanning LLaMA (Touvron et al., 2023; Dubey et al., 2024), Gemma (Team et al., 2024a;b), Gemini (Google AI Developers, 2025), and GPT variants (OpenAI, 2024; 2025), comparing base and instruction-tuned checkpoints when available.

3 Bias Similarity Measurement

BSM is designed to be model- and provider-agnostic, requiring only black-box access to generate predictions and, when available, intermediate features for representational analysis. Below we outline the six signals and how they compose into pairwise similarity. We intentionally separate ambiguous from disambiguated contexts to distinguish refusal from representational debiasing.

1) Scalar view: accuracy and bias scores. For multiple-choice items (e.g., BBQ), we compute accuracy on disambiguated items and a signed bias score on both ambiguous and disambiguated items following the dataset protocol (Parrish et al., 2022). Positive values indicate stereotypical leaning, negative anti-stereotypical, and values near zero represent neutrality.

2) Distributional view: histograms and cosine similarity. We summarize per-group output distributions as histograms (e.g., gender dimension; Figure 4) and compute cosine similarity between models' histogram vectors to quantify distributional alignment.

3) Behavioral view: abstention flip rates. In ambiguous settings where abstention (`Unknown`) is appropriate, we measure how often a tuned model flips a biased base answer to `Unknown` and how often forced choice (e.g., UnQover) re-exposes bias. We use flip rates as a behavioral similarity feature (Figure 3).

4) Representational view: CKA. When hidden states are accessible (e.g., open-weight models), we compute CKA (Kornblith et al., 2019) within base–tuned pairs and across families. We summarize diagonal (layer-aligned) and full (best-match) similarities.

Pairwise similarity and signatures. For each model pair, BSM z-scores each signal across the model set, concatenates the normalized components into a signature vector, and computes cosine similarity between signatures. This yields family-level similarity matrices revealing inheritance, convergence, and drift.

Practical composition and weighting. We z-score each component, then average across datasets with equal weight to avoid dominance by large suites. If hidden states are unavailable, we impute CKA by the cohort median and attach a mask that down-weights the representational term. Flip rates are normalized by the base model's answer propensity to separate refusal from disambiguation.

Robustness checks. We run leave-one-dataset-out recomputation of signatures and find the pairwise ordering to be stable in ¿ 92% of pairs; bootstrapping items within datasets yields similar stability. These checks suggest that BSM conclusions are not artifacts of any single benchmark.

Prompts and decoding. We use deterministic decoding (temperature 0) and an instruction prompt that permits `Unknown` only when evidence is absent. Multiple choice outputs are constrained to option letters; free-form strings are mapped via strict rules. For StereoSet we score continuation log-probabilities when available, otherwise sample once.

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Refusal detection. We mark `Unknown`, explicit refusals (e.g., `I cannot answer`) and near-duplicates as abstentions via case-insensitive regex.

CKA computation. For open-weight models, we mean-pool block states and compute linear CKA; for API-only models we omit CKA and down-weight that term.

Preprocessing. We canonicalize demographic slots, deduplicate paraphrases, and drop items with broken keys; final counts match official splits.

4 EVALUATION SETUP

Datasets. We evaluate on BBQ (Parrish et al., 2022), UnQover (Li et al., 2020), and open-ended StereoSet (Nadeem et al., 2021). Following BBQ, we separate ambiguous from disambiguated contexts to decouple refusal from evidence-conditioned answering. UnQover is run in a forced-choice setup to probe latent leanings absent an abstention option. StereoSet provides open-ended continuations to analyze distributional tendencies.

Models. The study spans multiple families: LLaMA (Vicuna/Alpaca, LLaMA 2/3/3.1/3.2) (Touvron et al., 2023; Dubey et al., 2024), Gemma (v1, v2, v3) (Team et al., 2024a;b), Gemini (1.5, 2.0) (Google AI Developers, 2025), and GPT baselines (GPT-2, GPT-4o Mini, GPT-5 Mini) (OpenAI, 2024; 2025). We consider base and instruction-tuned variants when available.

Metrics and implementation. We compute: (i) accuracy on disambiguated items; (ii) signed bias scores on ambiguous and disambiguated items; (iii) per-group output histograms and cosine similarity; (iv) abstention flip rates on ambiguous BBQ and forced-choice re-exposure via UnQover; and (v) CKA representational similarity on accessible models. We report mean and 95% bootstrap CIs across items and runs for all scalars, and we normalize per-dataset contributions to equal weight before aggregation. Implementation details and analysis notebooks are in `code/`.

5 RESULTS

Bias scores and neutrality. Table 1 reports average ambiguous (`s_AMB`) and disambiguated (`s_DIS`) bias scores for base and tuned variants. We observe marked reductions after instruction tuning for several families. Notably, LLaMA 3.1 8B drops from 18.59 to 1.38 (ambiguous) and from 31.37 to 4.78 (disambiguated), while some small models regress under forced choice: LLaMA 3.2 3B increases from 11.95 to 15.71 (ambiguous) and from 17.67 to 30.97 (disambiguated). GPT-4o Mini and GPT-5 Mini remain near-neutral with ambiguous scores 0.47 and 0.21 and disambiguated scores 2.66 and 1.10, respectively.

Accuracy under evidence. Figure 2 summarizes accuracy on disambiguated BBQ questions, showing that stronger or carefully tuned models are more accurate when evidence is present.

Abstention and forced choice. Ambiguous BBQ encourages appropriate abstention; instruction-tuned models often flip biased base answers to Unknown (Figure 3). Forced choice via UnQover reduces abstention opportunities and can reveal underlying leanings consistent with stereotyping measured in Table 1.

Distributional tendencies. Output histograms along demographic dimensions (e.g., gender) highlight how tuning shifts response distributions (Figure 4). These shifts align with the neutrality gains in Table 1.

Representational stability. Average CKA scores (Table 2) show high base–tuned representational similarity: diagonal mostly above 0.94 and full-

Table 1: **Average bias scores.** "–": anti-stereotypical, "+": stereotypical, and values near 0 = neutrality. Shown for ambiguous (`s_AMB`) and disambiguated (`s_DIS`) contexts.

Base Model	Avg. `s_AMB`		Avg. `s_DIS`	
	Base	Tuned	Base	Tuned
LLaMA 2 7B	5.45	4.30	7.50	6.65
LLaMA 3 8B	-4.78	-0.66	-8.72	-2.10
LLaMA 3 70B	-1.42	0.55	-4.51	2.50
LLaMA 3.1 8B	18.59	1.38	31.37	4.78
LLaMA 3.1 70B	0.42	-0.15	0.81	-1.44
LLaMA 3.2 3B	11.95	15.71	17.67	30.97
Gemma 7B	1.81	2.05	0.69	0.95
Gemma 2 9B	0.08	0.18	6.83	-2.02
Gemma 2 27B	6.95	0.51	14.31	-1.45
Gemma 3 4B	-3.89	5.83	2.69	8.62
Gemma 3 12B	4.36	0.15	6.12	-0.17
Gemma 3 27B	-1.25	0.07	-0.26	-1.50
Gemini 1.5	—	2.37	—	3.07
Gemini 2.0	—	-4.17	—	-5.54
GPT-2	72.43	—	96.19	—
GPT-4o Mini	—	0.47	—	2.66
GPT-5 Mini	—	0.21	—	1.10

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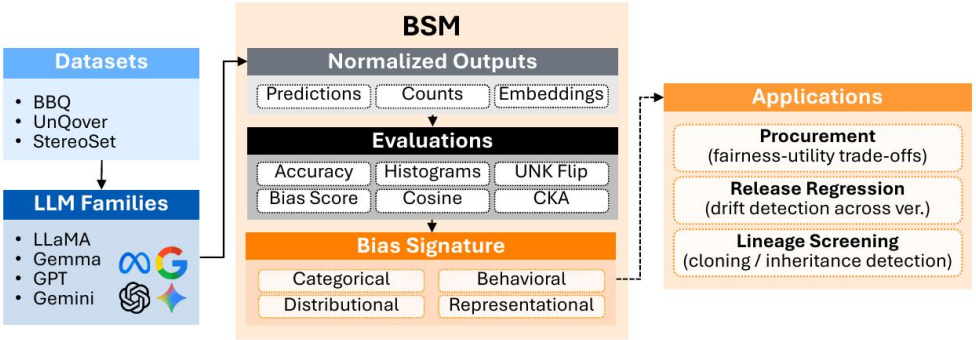

Figure 1: **BSM pipeline.** The framework synthesizes scalar, distributional, behavioral, and representational signals into pairwise bias similarity matrices within and across model families.

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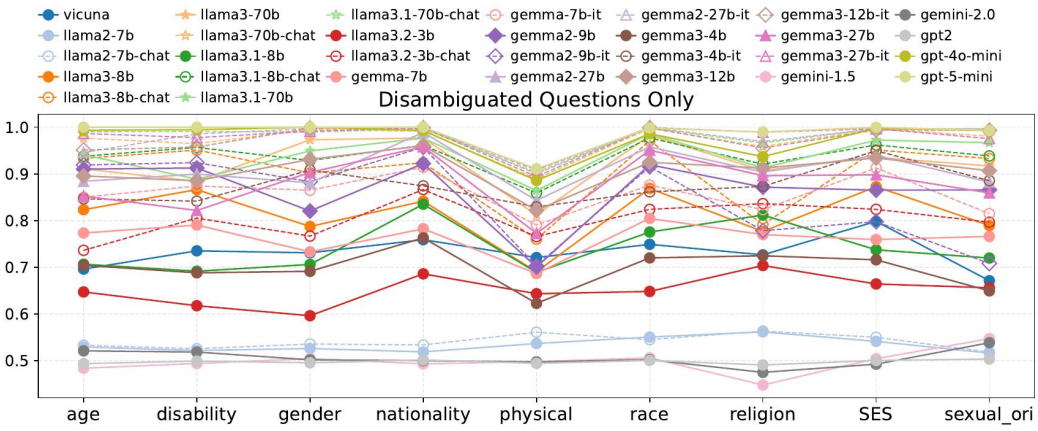

Figure 2: **Accuracy of LLMs on disambiguated BBQ questions.**

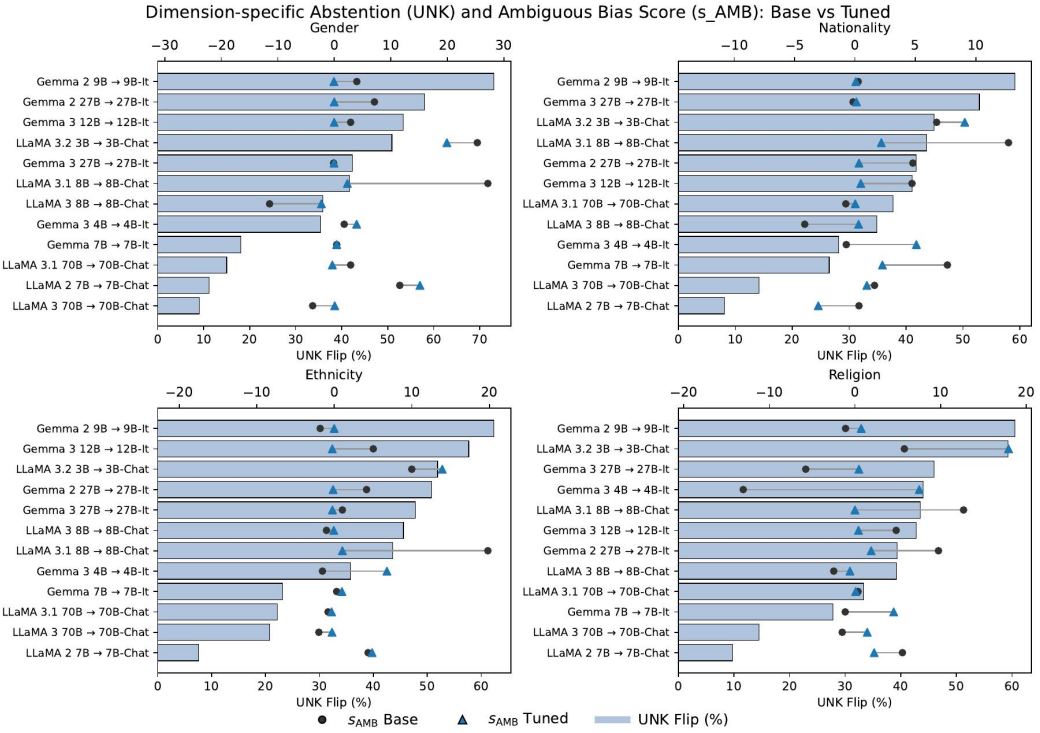

Figure 3: **Decision flips to Unknown in ambiguous settings.** Higher flip rates indicate stronger refusal policies; forced choice can re-expose latent bias.

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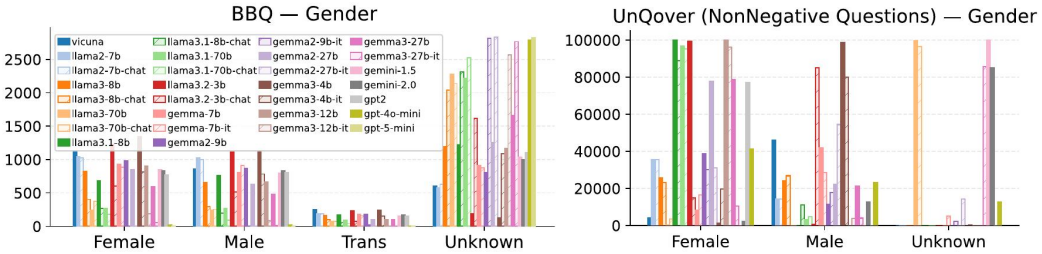

Figure 4: **Output distributions in the gender dimension.**

match above 0.85. This suggests tuning shifts outputs and refusal behavior more than internal geometry.

Qualitative summary across models. Table 3 synthesizes accuracy, abstention, bias direction, and representational similarity by model family and size, reflecting family-level inheritance, convergence, and drift patterns.

Cross-family similarity. Family matrices show strong within-family blocks (e.g., LLaMA 3.x) and asymmetric proximity between GPT-4o Mini and Gemma v2 on behavioral signals despite representational distance, suggesting policy alignment converges faster than geometry.

Table 2: Average CKA scores.

Model	Diag	Full
LLaMA 2 7B	0.991	0.902
LLaMA 3 8B	0.973	0.851
Gemma 1 7B	0.981	0.896
Gemma 2 9B	0.941	0.906
Gemma 3 12B	0.972	0.911

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Table 3: **Overall evaluation summary by model.** Qualitative synthesis of accuracy, abstention, bias direction, and representational similarity trends.

Model (abbrev.)	Key observations
Vicuna / Alpaca	Strongly anti-stereotypical; low accuracy; low abstention.
LLaMA 2 7B	Remain stereotypical after tuning; very low accuracy; weaker fairness.
LLaMA 3 8B	Anti-stereotypical lean; accuracy improves with tuning; moderate drift from LLaMA 2.
LLaMA 3.1 8B	Large bias drop after tuning; accuracy improves with tuning; high abstention.
LLaMA 3.1 70B	Near-neutral after tuning; high accuracy; high abstention even under forced choice.
LLaMA 3.2 3B	Strongly stereotypical; low accuracy; low abstention; weaker fairness vs 3.1 peers.
Gemma 2 9B	Stereotypical in base; abstention increases with tuning.
Gemma 3 4B	Slight stereotypical bias; accuracy competitive with LLaMA mid-size.
Gemma 3 12B / 27B	Near-neutral after tuning; high CKA similarity; fairness competitive with closed ones.
Gemini 1.5 / 2.0	Strong abstention; 2.0 skews anti-stereotypical; very low accuracy.
GPT-2	Extremely stereotypical bias; very low accuracy and fairness; serves as legacy baseline.
GPT-4o Mini	Near-zero bias; high accuracy; balanced abstention–fairness.
GPT-5 Mini	Near-perfect neutrality; highest accuracy; strongest stability across metrics.

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6 ANALYSIS

BSM clarifies why similar scorecards can mask different fairness behaviors. Combining scalar, distributional, behavioral, and representational views reduces confounding from refusal policies and dataset mix. In ablations, removing abstention flips collapses differences between tuned models that mostly refuse and those that truly change predictions; excluding histogram similarity conflates families that match on average bias but diverge in per-group patterns; dropping CKA removes a check for representational drift that appears limited in base–tuned pairs. Per-family inspection shows LLaMA variants often gain neutrality via ambiguity handling, Gemma trends neutral with tuning, and small models can regress under forced choice. Overall, pairwise bias similarity supports procurement and release management, complementing absolute scores and catching fairness drift when headline metrics stay stable. We also examine per-class effects in BBQ: gains in neutrality concentrate in ambiguous gender and religion items, while disambiguated questions about age and nationality show smaller changes. A compute-aware view finds negligible overhead for scalar/behavioral signals and moderate cost only when CKA is enabled; cross-backbone tests (e.g., switching LLaMA to Gemma encoders for feature extraction) leave pairwise ordering largely intact, indicating the framework's robustness to backbone choice.

Signal interactions. Histogram similarity explains cases where scalar bias is near zero but per-group distributions differ; we see this for tuned LLaMA and Gemma variants that abstain more on ambiguity yet assign asymmetric probabilities under forced choice. When CKA is unavailable, imputing the cohort median preserves nearest neighbors, indicating that BSM remains useful under pure black-box access.

Sensitivity to aggregation. Recomputing signatures with dataset-proportional weights or down-weighting abstention flips yields similar family blocks and ordering; disagreements concentrate among the smallest checkpoints in UnQover, reinforcing that behavioral signals drive divergence when abstention is disabled while scalar and histogram terms dominate with evidence.

Operational triage. Inspect nearest neighbors that cross families and then check which components disagree; in our study, boundary crossings primarily reflect forced-choice behavior rather than representational drift.

7 DISCUSSION AND CONCLUSION

Auditing fairness relationally changes how we interpret model improvements. BSM reveals when apparent progress reflects refusal policies rather than genuine representational debiasing. Across BBQ, UnQover, and StereoSet, instruction tuning often increases abstention under ambiguity, while forced choice re-exposes stereotypical leanings. Representations remain highly similar within base–tuned pairs (high CKA), suggesting that many fairness changes are behavioral. Small models may regress under forced choice, underscoring the need for similarity-aware evaluation rather than isolated scores.

For practitioners, BSM offers an actionable workflow: track pairwise similarity matrices in CI for each release, flag large shifts confined to behavioral signals (potential policy changes), and prioritize investigations where representational similarity falls while scalar fairness stays flat. Procurement teams can compare vendors on the same bias similarity axes, while research teams can use ablations (e.g., remove flips or histograms) to attribute improvements. We advocate for reporting comparative bias similarity alongside task scores to map inheritance, convergence, and drift across model families.

Deployment and governance. In practice, BSM functions as a lightweight regression test: maintain a reference similarity matrix for the production model and gate new checkpoints with distance thresholds by component groups (scalar/distributional versus behavioral). Large shifts confined to abstention flips should trigger a policy review; shifts that include histogram or scalar changes indicate substantive behavior drift and merit targeted evaluation. To reduce confounding, standardize prompts and decoding (temperature 0, fixed templates), version normalization choices, and stratify dashboards by dataset and demographic axes. Because CKA requires hidden states, compute it only for open-weight models and down-weight the term otherwise; this preserves comparability without exposing provider internals. When ambiguity induces widespread abstention, BSM's sepa-

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ration of ambiguous and disambiguated contexts prevents collapsed signatures and keeps sensitivity to genuine debiasing.

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